



Automatic Extraction of Electoral Results from the Tally Sheets of the 2026 Peruvian General Elections

Josemanuel Rossy Cañari Palante · Kenny Asto Hinostroza · Melissa Dessire Aylas Barranca · Carlos Pérez Pérez

Master's Program in Artificial Intelligence — Computer Vision, Group 3 — Universidad Nacional de Ingeniería, Lima, Peru



code + data

59.6%

field accuracy on handwritten actas
95% CI [56.6, 62.5] · CER 0.401 · 3,268 fields

95.1%

on born-digital actas, parsed from the PDF text layer **with no vision at all**

0

manually annotated fields: ground truth is the official result published by the same portal

1. Problem

Peru's National Office of Electoral Processes (ONPE) publishes every tally sheet (*acta de escrutinio*) digitally, but the information is **visual**: a scanned form where polling-station members wrote the vote counts by hand. Manual extraction does not scale to the ~88,000 stations of a general election, which limits automatic auditing and independent verification. **Our question**: how far can a classical computer vision pipeline go on *real, nationally sampled* actas, and what does that imply for a production system?

2. Dataset and ground truth

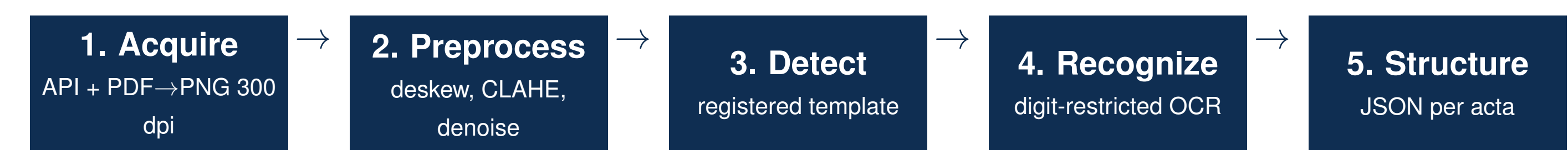
Source. Official ONPE portal, 2026 General Elections, first round (presidential); actas downloaded through the portal's own API.

Sampling. Simple random sample without replacement over the ~88,064 stations with a **published seed (2026)**: 100 actas, 100% hit rate, 23 department-level codes, size-proportional (Lima 36%). This removes the bias of the pilot, which used sequential codes from a single district.

Annotation-free ground truth. The same portal publishes the officially digitized counts, so every acta yields ~43 labeled field–value pairs **at zero annotation cost** and the evaluation is fully reproducible.

Population finding with architectural consequences. The population is *mixed*: **76 handwritten** scans and **24 born-digital (STAE)** documents with an extractable text layer (prevalence 24%, 95% Wilson CI [16.7, 33.2]). A production system must **classify the document type first and route accordingly**.

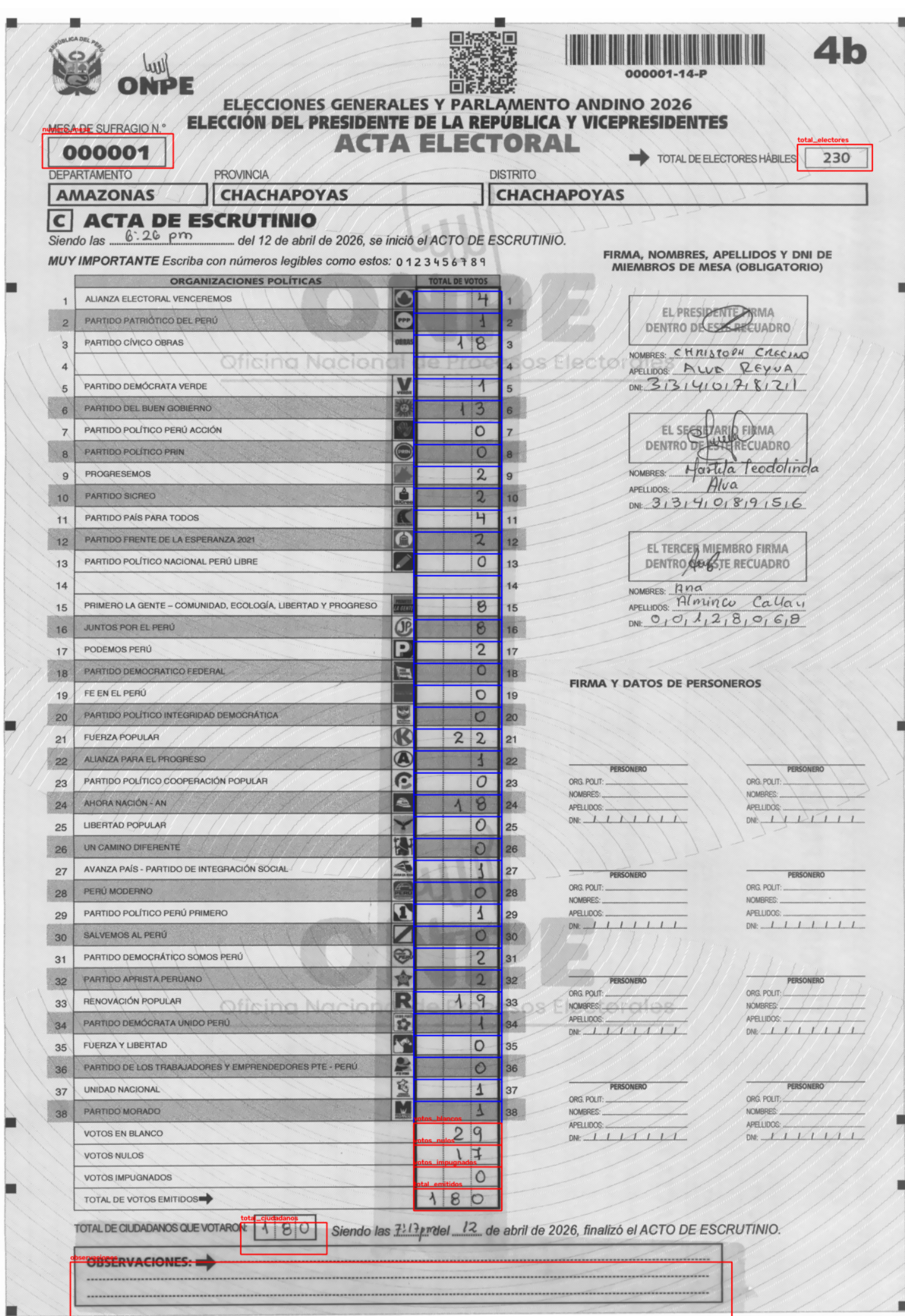
3. Method



Detection by fiducial registration. The acta is a fixed-layout form, so 46 regions (38 party rows, 4 summary rows, 4 context fields) are defined as page fractions. Real scans break fixed fractions (black bands, offsets), so the 15 printed registration squares are detected, matched to a reference acta by **RANSAC on translation**, and corrected with a partial affine warp before cropping.

Recognition. EasyOCR restricted to digits over cleaned cells: interior margin against printed borders, ink isolation by dark threshold and Otsu, erasure of the dotted separators by template position, speckle filtering.

Domain rule. Polling-station members leave a cell *blank* instead of writing 0, so an empty cleaned cell is read as **0 votes** — a semantic decision, not a vision one.



Left: template regions after fiducial registration. Right: cell cleaning (raw crop → ink isolation → separators erased).

4. Experimental setup

Metrics. *Field accuracy* (exact match against the official integer) and *CER* over digit strings.

Statistics. Fields inside one acta share writer and scan quality, so **the acta is the cluster**: every interval comes from a **clustered bootstrap** (10,000 resamples of whole actas). Per-field binomial intervals would be optimistic — the measured design effect is **3.07** on handwritten actas (76 actas ≈ 1,066 independent fields). Ablations are re-evaluated on the saved outputs, with no OCR re-run.

5. Results

National random sample — 76 handwritten actas, 3,268 fields

Configuration	Acc. (%)	CER
Fixed template, raw OCR	26.81	0.724
Fixed template + zero rule	50.86	0.482
Registered (fiducials), raw OCR	38.00	0.617
Registered + zero rule	59.58	0.401

Fiducial registration is worth **+8.7 points** and lifts the worst acta from 14% to 30%; the domain rule is worth **+21.6 points**. Per-acta accuracy averages 59.6% (SD 13.0, range 30–84%): the residual variance is handwriting and scan quality, not misalignment.

Pilot ablation — 10 actas, 430 fields

Configuration	Acc. (%)	CER
v1: raw digit OCR	43.95	0.524
v2: + ink isolation, borders, sub-boxes	52.09	0.419
v3: + separator erasure, Otsu, speckle	52.79	0.476
v4: + empty-cell-is-zero rule	58.37	0.420

Does a multimodal LLM replace the pipeline? No. Same 15 handwritten actas (645 fields), same protocol; the LLM gets the full page plus two image–JSON exemplars, with no template and no OCR.

Method	Acc. (%)	CER
Pipeline (registered template + OCR)	60.93	0.395
LLM few-shot, end-to-end (2 exemplars)	48.37	0.591

The LLM reads the layout well but is not trained for handwritten Peruvian digits and collapses on poor scans (20.9% and 23.3% on the two worst actas, where the pipeline still reads 34.9% and 53.5%). Its real advantage is **zero detection engineering**: a bootstrap pre-labeler for a new form, not a replacement recognizer.

Is 24 actas enough for the 95.1%? Yes — text parsing is deterministic, so between-acta variance falls below binomial ($ICC \approx 0$, design effect 1.0) and the interval is [94.5, 95.7], nine times tighter than the handwritten one. The 35.5-point gap has a 95% CI of [32.6, 38.5].

6. Conclusions and future work

- Register the template on the form's fiducial marks *before* any fixed-coordinate cropping.
- Separate domain semantics (blank = 0) from vision performance; conflating them hides where the system really fails.
- Classify the document type first: routing born-digital actas to text parsing is worth **35.5 points**.
- Generality does not yet beat specialization: a general multimodal LLM is cheaper to build but 12.6 points less accurate.

Limitations. 59.6% is a *ceiling for generic OCR*, not a deployable accuracy. Errors now concentrate in **recognition**, not localization — exactly the component fine-tuning replaces. Next: fine-tune TrOCR on the free field–value pairs (~43,000 from 1,000 actas), train YOLOv11 / RT-DETR on labels auto-generated by the registered template, and use the LLM as pre-labeler in the human-correction loop.